

Competitive Exam Practice 5: Addressing Modes, CPU Organization, and Bus Structures

GATE (Computer Science) Pattern

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type.

Q1 [*GATE CS 1996 Q1.16, verified — GATEOverflow V3 p.29*] [**MCQ, 1 mark**] Relative mode of addressing is most relevant to writing:

- (A) Co-routines
- (B) Position-independent code
- (C) Shareable code
- (D) Interrupt handlers

Q2 [*GATE CS 1998 Q1.19, verified — GATEOverflow V3 p.29*] [**MCQ, 1 mark**] Which of the following addressing modes permits relocation without any change whatsoever in the code?

- (A) Indirect addressing
- (B) Indexed addressing
- (C) Base register addressing
- (D) PC-relative addressing

Q3 [*GATE CS 1999 Q2.23, verified — GATEOverflow V3 p.29*] [**MCQ, 2 marks**] A certain processor supports only the immediate and the direct addressing modes. Which of the following programming-language features cannot be implemented on this processor?

- (A) Pointers
- (B) Arrays
- (C) Records
- (D) Recursive procedures with local variables

Q4 [*GATE CS 2000 Q1.10, verified — GATEOverflow V3 p.29*] [**MCQ, 1 mark**] The most appropriate matching for the following pairs is:

X: Indirect addressing	1: Loops
Y: Immediate addressing	2: Pointers
Z: Autodecrement addressing	3: Constants

- (A) X-3, Y-2, Z-1
- (B) X-1, Y-3, Z-2
- (C) X-2, Y-3, Z-1
- (D) X-3, Y-1, Z-2

Q5 [*GATE CS 2002 Q1.24, verified — GATEOverflow V3 p.30*] **[MCQ, 1 mark]** In the absolute addressing mode:

- (A) the operand is inside the instruction
- (B) the address of the operand is inside the instruction
- (C) the register containing the address of the operand is specified inside the instruction
- (D) the location of the operand is implicit

Q6 [*GATE CS 2004 Q20, verified — GATEOverflow V3 p.30*] **[MCQ, 1 mark]** Which of the following addressing modes are suitable for program relocation at run time?

- (A) Absolute addressing only
- (B) Based and relative addressing
- (C) Indexed addressing only
- (D) Indirect addressing only

Q7 [*Original, GATE-pattern*] **[MCQ, 2 marks]** In a single-bus CPU, the instruction $R1 \leftarrow R2 + R3$ is implemented over three control steps using two hidden registers. The reason the same transfer takes only *one* step on a three-bus organization is that:

- (A) the ALU is faster on a three-bus design
- (B) three buses can carry both operands and the result in the same cycle without a shared-path bottleneck
- (C) the register file has more registers on a three-bus design
- (D) control logic is simpler on a three-bus design